VanniKawachh: A Distributed AI Acoustic Intelligence and Autonomous Drone Response Network for Women's Safety

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***Abstract*—Women are particularly at risk in areas with poor mobile coverage, parking lots, dark streets, and woodland trails where standard safety equipment breaks down. The majority of current options rely on a panic button, wearable technology, a phone, or an application. These restrictions may cause assistance to be delayed during an emergency. VanniKawachh shifts accountability from the victim on public infrastructure. Installed in high-risk areas, solar-powered sensor nodes listen for screams, cries, or utterances like "Bachao" and "Help." Before an alarm is sent, the first step detects a potential distress event close to the microphone; the second stage verifies the sound together with motion, light, and time data. The solution does not rely on a SIM card or mobile data because a validated alert transmits the measured position via an encrypted LoRa channel. The alarm can start an autonomous drone mission to monitor the area, deliver a first-aid kit, and assist responders. It is shown on a police dashboard. The system design, module-level implementation, held-out audio-classifier results, and software-in-the-loop flight validation are all presented in this study. The goal is straightforward: a woman's voice should be able to attract attention and help without the need for a safety gadget.**

***Index Terms*—Women's safety, acoustic event detection, MFCC, convolutional neural network, PANNs, edge computing, LoRa, AES encryption, autonomous quadcopter, Pixhawk, public safety infrastructure.**

I. Introduction

Someone seeking help may be unable to reach a phone or press a panic button. The phone may be missing, its battery may be empty, or the area may have no mobile signal. VanniKawachh asks whether a fixed sensor at the location could hear a call for help and alert responders without requiring the person to carry anything.

The proposed network places a microphone and small computer on a public pole. When the node hears a possible scream, cry, or call for help, it sends a short alert to a nearby hub. The hub checks a brief recording and the surrounding conditions before showing the incident to an operator or requesting a drone mission. A low-bandwidth LoRa link carries the alert; a local Wi-Fi link carries the audio clip. The hub uses its record of the pole's installed location to locate the incident.

A mistaken sound should not launch an aircraft, and a real call should not be ignored. For that reason, the system makes the decision in stages and checks the vehicle before flight. This paper reports an engineering prototype: the audio classifier was evaluated on held-out recordings, while the complete flight sequence was tested with a simulated aircraft. The photograph in this paper shows a supervised hover, not a fully automatic field response. Section II reviews related work, Section III explains the system modules, and Section IV presents the measured results and their limits.

II. Literature Survey

Victim-carried systems. Pavithra *et al.* propose on-phone audio recognition and emergency automation [1]; Potturi *et al.* use a GSM/GPS SOS device [2]. Both shorten the alert workflow but still require a personal device. Bharathi *et al.* report 97.54% accuracy for a wearable using GPS and acceleration [3]. The button and wearable systems of Snehith *et al.*, Hadkar *et al.*, and Uganya *et al.* likewise depend on a worn or reachable trigger [4]–[6]. Their reported metrics describe different tasks and datasets, so they cannot be compared directly with acoustic recall in this paper.

Acoustic detection. A smart-city sound-event review identifies the breadth of existing methods [7]. Kim *et al.* study CNN–Transformer scream detection and temporal localization [8], following their larger scream-dataset study [9]. Sharma and Jebaseeli report 92% accuracy for a scream/panic detector [10]; Srimathi *et al.* examine scream analysis for surveillance [11];

TABLE I
RELATED WOMEN-SAFETY SYSTEMS.

Sr. No.	Title / Reference	Year	Algorithm / Technique	Accuracy / Result	Remark
1	AI-powered women's safety app [1]	2026	Smartphone audio recognition and automated emergency actions	Real-time distress recognition; accuracy not reported	Requires the victim's smartphone.
2	SOS women safety device [2]	2026	GSM, GPS, automatic SOS and location sharing	Emergency alert and geo-location sharing	Depends on GSM availability.
3	Sound-event detection in smart cities [7]	2026	Review of ML/DL sound-event detection methods	Survey of datasets, methods and applications	Detection-focused; no response system.
4	TinyML urban sound classification [13]	2026	TinyML and adaptive edge classification	Accuracy improved from 78.27% to about 85%	Supports edge acoustic inference.
5	IoT-enabled women safety system [3]	2025	IoT sensors, GPS, accelerometer and microphone	Reported emergency detection and alerting	Requires a wearable device.
6	Smart IoT safety device [4]	2025	ESP32, GPS/GSM, push button and sensors	Emergency location and alert transmission	Device-dependent safety mechanism.
7	CNN–Transformer scream detection [8]	2025	CNN–Transformer and Windowing CNN	Improved F-measure and equal-error rate	Detection without physical response.
8	Smart scream and panic detection [10]	2025	AI-based scream and panic recognition	Reported scream and panic detection	Alerting remains device/network dependent.
9	Human scream detection for crime reduction [11]	2025	ML-based scream detection and alerting	40% lower reaction time; 30% higher intervention	No autonomous physical response.
10	Drone-delivered AED response study [19]	2025	Autonomous drone dispatch and payload delivery	Successful delivery in 9 of 11 simulations	Supports drone-based first response.
11	Efficient IoT women safety device [5]	2024	IoT-based personal safety and location tracking	Prototype demonstrated	Requires a personal device.
12	Audio danger detection [12]	2024	Noise reduction and deep-learning classification	Noise-aware danger-sound classification	Mainly addresses detection accuracy.
13	IoT and GPS safety tracker [6]	2023	Manual trigger, GPS and GSM	Location tracking and emergency alert	Manual trigger and GSM dependency.
14	LoRa emergency management system [18]	2020	LoRa emergency communication and localization	Emergency communication demonstrated without normal infrastructure	Supports offline alert transmission.

and Fime *et al.* report 95.51% danger-sound accuracy with InceptionV3 [12]. These papers motivate acoustic sensing, while VanniKawachh examines what happens after a candidate sound is detected.

Enabling work. TinyML studies support compact edge inference [13], [14]. PANNs and AudioSet provide a hub-side audio-tagging baseline [15], [16]; MFCCs remain a compact speech representation [17]. A LoRa emergency-system study addresses low-rate alerting [18], and a simulated medical-drone study shows the potential of aerial response [19]. The implementation also draws on the AES and HMAC specifications [20], [21], Semtech radio documentation and the LoRaWAN specification [22], [23], and ArduPilot/MAVLink interfaces [24]. Deployment considerations include India's data-protection law, drone rules, and radio notifications [25]–[27]. Table I retains the journal paper's comparison, with its outcomes interpreted only within each cited study.

III. METHODOLOGY AND SYSTEM ARCHITECTURE

Fig. 1 outlines the intended sequence. The fixed node is designed around an ESP32-S3 with an INMP441 microphone, PIR sensor, light sensor, solar supply, and LoRa radio. Commissioning records the node's position in a hub registry. A candidate event sends a sealed alert over LoRa and a four-second clip over local Wi-Fi. The figure's tracking step remains future work; it was not part of the measured response chain.

A. Acoustic Screening and Verification

The trained Stage-1 int8 model uses MFCC features for background, scream, cry, and help classification. Its 34 kB size and held-out results are measured offline; ESP32 latency and outdoor accuracy remain unmeasured. The hub waits for a clip and runs a configured verifier. The intended Pi route uses PANNs; the deployable fallback uses YAMNet and related acoustic checks. The current full-chain SITL rehearsal used a labelled development fallback. In the current hub pipeline, a missing clip is logged without dispatch; it is not replaced by a Stage-1 score.

The node and hub have different jobs. A low threshold on the node creates a candidate; it does not authorize a flight. The node packages the event type, confidence, PIR reading, light level, and battery state while retaining the short audio window for upload. The hub matches the clip to the node identifier and event counter in the alert, and only then performs the second audio decision. This pairing prevents a late clip from being treated as evidence for a newer incident. The trained model's offline evaluation does not yet establish performance with the INMP441 in an outdoor installation.

For a verified clip, the hub combines its audio score a with normalized Stage-1 confidence c, PIR motion p, darkness $d = 1 - L/255$, and a night indicator n:

$$S = 0.60a + 0.15c + 0.10p + 0.08d + 0.07n. \quad (1)$$

The weights are prototype policy values, not learned probabilities. The live pipeline requires an affirmative Stage-2 confirmation and $S \geq 0.60$ before dispatch. The incident is logged even when the dispatch gate fails. A separate voice-distress TFLite model for quiet and muffled speech is specified in the repository but has not yet been trained.

The sound verification result and the fused score are separate gates. A high motion reading or a dark street can increase priority after audio confirmation, but cannot turn an

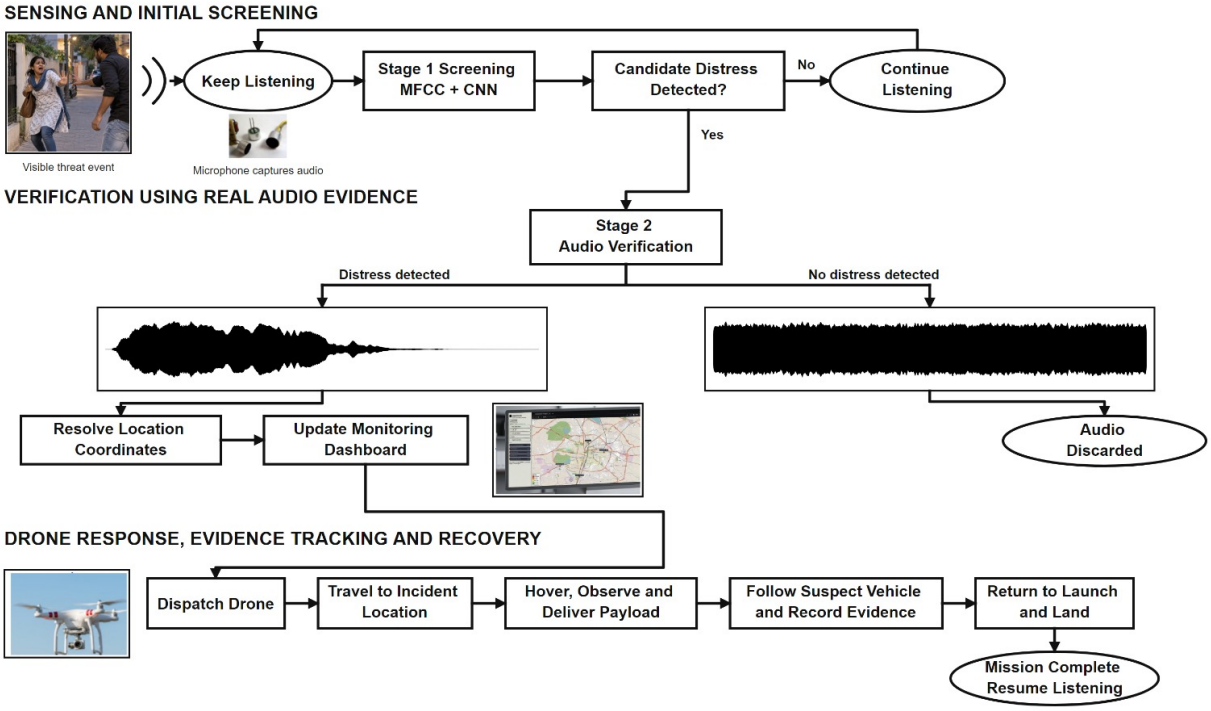


Fig. 1. Author-supplied conceptual workflow from sensing through response. In the implemented pipeline, the node sends its alert and audio clip on separate links, the hub resolves the registered node's location, and dispatch requires Stage-2 confirmation. Suspect-vehicle following is a proposed extension, not a measured result.

unconfirmed clip into a dispatch. A missing clip leaves the second gate unresolved. This is conservative for flight safety, although it also means that a real shout may be missed when Wi-Fi fails. Measuring that trade-off requires paired outdoor audio and link-loss trials; the held-out classifier set alone cannot estimate it.

Fig. 2 illustrates why the acoustic task is more than loudness detection. A genuine event can be quiet, distant, or muffled; laughter and excited speech can also be loud and spectrally broad. The model must therefore be tested on both hard negatives and degraded positives before a field dispatch threshold is set.

B. Authenticated Alert and Dispatch

The 25-byte packet has a nine-byte clear header (magic, version, node identifier, monotonic counter), an eight-byte encrypted evidence payload, and an eight-byte truncated HMAC tag. The implemented code uses AES-128-CTR and a per-node key derived from a master secret. The hub verifies the tag before decryption and rejects stale counters and unknown nodes. The packet contains no coordinates: the hub resolves the node identifier to surveyed coordinates. The audio clip stays on the local Wi-Fi path, since a four-second PCM clip is much larger than the LoRa alert. Persistent counter handling, key separation, and field link reliability still require deployment validation.

This split makes the alert small enough for low-rate radio while keeping the verification recording off that channel. The gateway forwards packet bytes to the hub without making a dispatch decision. The hub first authenticates the packet, then checks the counter against the last accepted value for that node, and finally looks up the surveyed location. A registry miss, invalid tag, or repeated counter ends processing before a mission request is formed. For field use, counters must survive node resets and the master secret must be kept outside the captured field node; those properties have not been established by the SITL run.

C. Drone Response and Safety Controls

The dashboard and dispatcher pass a verified incident to the flight API. The response software checks the target geofence, serializes missions in a bounded priority queue, and requires the vehicle to be landed and disarmed before the next flight. Mode changes are retried through MAVLink until the autopilot's reported mode confirms them. The failsafe arbiter evaluates battery, GPS, link, geofence, and progress; LAND takes precedence over RTL. The SITL mission includes takeoff, transit, hover, a simulated first-aid release, and return.

The mission executor moves through connection, GPS readiness, arming, takeoff, transit, hover, delivery, return, and landing states. Each phase checks the arbiter rather than relying on a single pre-flight assessment. A brief invalid GPS report is debounced; a persistent loss requests LAND because a GPS-dependent return is no longer reliable. Battery escalation from low to critical also changes the requested recovery from RTL to LAND. The queue worker checks disarm before beginning another mission, so a second incident cannot start a new takeoff

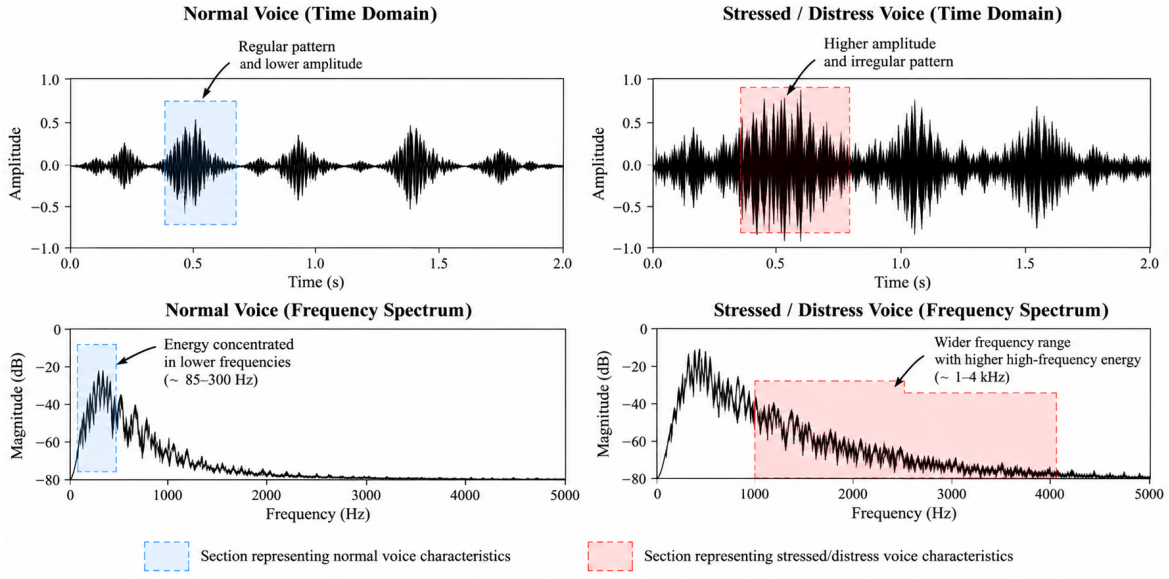


Fig. 2. Author-supplied illustrative waveforms and spectra for ordinary and stressed vocalizations. The curves are a conceptual comparison, not recordings from the study or a calibrated rule: amplitude, pitch, and high-frequency energy alone cannot establish distress.



Fig. 3. Author-supplied photograph of the assembled quadcopter during supervised hover. The image documents the airframe, not an end-to-end autonomous flight or a measured payload-delivery result.

while the first vehicle remains airborne. These are software safeguards layered over the autopilot's own configuration, not evidence of physical flight safety.

Fig. 3 shows the physical airframe. Current operator-controlled real-mode gates remain disabled pending manual outdoor validation, GPS and battery readiness, and microphone false-alarm testing. The flight photograph should therefore be read separately from the SITL performance reported below.

The proposed delivery sequence descends from observation height, releases the kit only near the ground, then climbs and returns. The release command is recorded as an outcome: a failed actuation must be visible to the operator, but it must not hold the aircraft above the incident indefinitely. The control software can exercise that sequence in simulation; actual drop accuracy, bystander clearance, and camera tracking have not been measured here.

IV. Implementation and Results

Table II compares the related approaches with the proposed response chain. The studies used different data, so their accuracy figures are not directly comparable.

A. Offline Audio and Software Evaluation

The held-out Stage-1 evaluation used 1,719 clips; Table III gives class-wise results. Overall accuracy was 86.62%, with macro F1 of 0.79 and weighted F1 of 0.86. The scream recall of 0.56 means that 44% of labelled screams in this split were missed. The 14 help clips are too few to support a broad language or field claim. The reported result applies to the offline model and cannot be used as a false-dispatch rate for the complete system.

The ArduCopter SITL acceptance run passed eight of eight checks, including arming, target arrival, hover, RTL, and landed/disarmed completion. It reached 0.4 m closest approach against a 5 m arrival tolerance and took 328 s from trigger to disarm. In a separate full-chain rehearsal, a sealed alert passed registry lookup and fusion (severity 0.88), the simulated aircraft descended to about 3.1 m, a servo release was commanded, and the mission ended in COMPLETED. The rehearsal used heuristic acoustic stand-ins. The original 68 unit verifications passed for packet rejection, fusion, queueing, geofence, and flight-safety behaviour; the current repository has added further tests, so 68 is a historical verified count rather than a claim about the current total.

B. Evaluation Analysis

Reading the acoustic result. The test split is dominated by background clips (1,386 of 1,719). This makes overall accuracy a poor stand-alone measure of emergency usefulness:

TABLE II
COMPARISON WITH RELATED SYSTEMS.

Method used	Existing system	VanniKawachh system	Advantages of VanniKawachh
IoT sensors, GPS, accelerometer, and microphone [3]	A wearable device detects a possible emergency and sends an alert.	Fixed sensing nodes use acoustic screening with motion, light, and time information.	Protection is available at the location without requiring a wearable device.
Smartphone audio recognition and emergency actions [1]	A smartphone application identifies possible distress and starts emergency actions.	A fixed acoustic node performs local screening and forwards a candidate event to the hub.	The user does not need to access, unlock, or operate a phone.
GPS, GSM, automatic SOS, and location sharing [2]	GPS obtains the position and GSM transmits the emergency message.	The hub maps a registered node ID to its installed location and receives alerts over LoRa.	The alert can work where mobile or GSM connectivity is unavailable.
Machine-learning sound-event detection [7]	Machine-learning methods identify sound events in smart-city environments.	Audio detection starts a larger chain containing verification, fusion, and dispatch logic.	Connects sound-event detection with an emergency-response workflow.
CNN–Transformer and window-based scream detection [8]	Deep-learning and temporal windows detect screams in recorded audio.	The node screens short windows, while the hub verifies the received event clip.	Separates recall-oriented screening from precision-oriented confirmation.
Noise preprocessing and deep-learning danger-sound classification [12]	Preprocessing and deep learning classify danger sounds in noisy environments.	The hub combines acoustic evidence with Stage-1 confidence, motion, darkness, and time.	Environmental context reduces decisions based only on a loud sound.
TinyML and adaptive edge classification [13]	Sound classification is demonstrated on microcontroller-class hardware.	An ESP32-S3 performs lightweight MFCC and CNN Stage-1 screening.	The first decision occurs near the microphone, reducing latency and communication needs.
LoRa emergency communication and localization [18]	LoRa transmits emergency information where regular mobile infrastructure is unavailable.	LoRa carries an authenticated alert, while a short-range link transfers the audio clip.	Supports low-bandwidth offline alerting with replay and spoofing protection.
Autonomous drone emergency payload delivery [19]	An autonomous aircraft delivers an emergency payload after receiving a request.	Drone dispatch occurs after alert authentication, audio verification, evidence fusion, and safety gating.	Links aerial response to a verified event with flight-safety interlocks.

TABLE III
HELD-OUT STAGE-1 ACOUSTIC CLASSIFIER RESULTS

Class	Precision	Recall	F1-score	Support
Background	0.92	0.93	0.92	1386
Scream	0.66	0.56	0.60	281
Cry	0.54	0.95	0.69	38
Help	0.88	1.00	0.93	14
Overall accuracy	0.866			1719
Macro F1-score	0.79			1719
Weighted F1-score	0.86			1719

the minority scream class is the one for which a missed event matters most. The model found only 56% of the 281 labelled screams. Cry recall was higher, but its support was 38 clips; the apparent perfect recall for help comes from just 14 clips. These counts limit the strength of any claim about unfamiliar voices, languages, recording distances, or street noise. The supplied spectrum illustration is helpful for explaining the problem, but no decision threshold or performance estimate was derived from that image.

Flight acceptance and failure paths. The SITL harness launched the flight stack and simulated ArduCopter, sent a request through the same HTTP interface used by the dispatch service, then read vehicle telemetry until landing. The eight checks cover acceptance, connection, arming, altitude, target approach, hover, RTL, and disarmed completion. The 0.4 m figure is the closest simulated approach to the requested coordinate, not a field localization error and not a first-aid drop error. The 328 s wall-clock duration includes the entire simulated mission, so it should not be presented as time to first contact. Unit checks separately exercise a corrupted packet tag, a stale replay counter, sub-threshold fusion, GPS debounce, LAND-over-RTL escalation, and refusal to start a queued mission against an armed vehicle. They test the implemented branches; they cannot establish that a damaged radio, uncalibrated sensor, or windy airframe will behave the same way.

Chain rehearsal. The sealed alert used a registry node and a development audio score to drive the downstream interfaces. The hub accepted the packet, resolved the stored coordinates, computed severity, and sent one mission request. The aircraft then followed the mission sequence in SITL. That rehearsal demonstrates that the modules can exchange the expected data and that the simulated mission can terminate cleanly. It does not evaluate the trained Stage-1 model and hub verifier together on the same live outdoor event. A defensible end-to-end detection rate will require one synchronized test set spanning the microphone, alert link, clip link, hub decision, and response outcome.

V. DISCUSSION

The present evidence supports a modular prototype and simulated flight chain. Field deployment would need a continuous-background false-alert rate, varied female speech and distress recordings, measured radio range and power, and physical flight tests with an operator. Audio is processed locally until a candidate event, but clips and camera evidence still require access control, retention rules, and notice. Physical UAS

operation requires review against current airspace and pilot rules; radio-band use must be checked against the applicable Indian spectrum notification. The prototype's cloud dashboard and phone path do not imply that a LoRa-only field node can reach an internet-hosted service without a local gateway.

The decision policy also has a human consequence. Refusing dispatch without a verification clip protects against a single noisy sensor launching an aircraft, but it cannot help someone whose alert packet arrives while the local clip link fails. A field study should record that failure rate and compare it with a supervised escalation path, such as sending an operator-visible incident without automatic flight. The hub is a locality-level dependency as well: losing it stops authentication, lookup, and dispatch even if pole nodes continue to hear sound. Redundant gateways and a defined recovery procedure belong in a deployment design, not in the performance claims of this prototype.

VI. Conclusion and Future Work

VanniKawachh connects fixed acoustic sensing, authenticated alerting, contextual verification, and a safety-interlocked drone response. Its offline classifier and SITL results establish separate technical baselines, while the complete field detection-to-flight rate remains unmeasured. The next evaluation should deploy the trained screen on the intended microphone, test hard negatives and muffled calls on unseen speakers, measure the two radio links, and then conduct controlled visual-line-of-sight flights. These measurements will determine whether the combined chain can support supervised emergency use.

Acknowledgment

The authors thank the Department of Computer Science and Engineering, G. H. Raisoni College of Engineering, Nagpur, for laboratory access and institutional support.

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